Chapter 1

Fisher information and the Cramér–Rao bound

The paper's information theorems are statements about Fisher information: how much one noised sequence tells an estimator about a coordinate of the clean law, and how that amount decays with the noise level. Its learning section then compares an estimator's dispersion with the Cramér–Rao reference built from that information, and its corollary on sample complexity invokes the local asymptotic minimax theorem of Hájek and Le Cam. This chapter defines the statistical score and keeps it apart from the diffusion score, derives Fisher information and its properties, proves the Cramér–Rao bound, works out the efficient information under nuisance parameters and why a nuisance at another Hermite degree costs nothing to leading order, derives the Gaussian information formula the paper uses in closed form, states local asymptotic normality and the Hájek–Le Cam theorem in plain words and in the form the paper's corollary uses, explains what "maximum likelihood attains it" does and does not say, and derives Fisher's identity for latent-variable models, which is how the paper computes the gradient of its marginal likelihood from one forward–backward pass.

1.1 Two scores

The word **score** has two meanings in this project, and confusing them makes the paper unreadable.

Definition 1.1 (Statistical score)**.** For a parametric family of densities $p_\theta(\mathbf{x})$ with $\theta \in \Theta \subset \mathbb{R}^d$, the **statistical score** at $\mathbf{x}$ is the gradient of the log-likelihood with respect to the *parameter*,

$$\mathbf{s}_\theta(\mathbf{x}) = \nabla_\theta \log p_\theta(\mathbf{x}) \in \mathbb{R}^d. \tag{1.1}$$

Definition 1.2 (Diffusion score)**.** For a density P_t on $\mathbb{R}^L$, the **diffusion score** (also Stein score) at $\mathbf{x}$ is the gradient of the log-density with respect to the *data*,

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) \in \mathbb{R}^L. \tag{1.2}$$

Both are gradients of a log-density; they differ in the variable of differentiation, in their dimension (d against L), and in their use. The statistical score is a random vector whose covariance is the Fisher information and whose sum over a sample is set to zero by maximum likelihood. The diffusion score is a vector field that drives the reverse process and that a network is trained to approximate. The paper writes "the score" for the diffusion score throughout, and "the likelihood score", "per-sequence scores obtained by Fisher's identity" or "$\partial_\theta \log P_t$" for the statistical one. The two meet in one place: the paper's Theorem 2 computes the Fisher information of a parameter of P_0 from the noised law P_t, so the *statistical* score $\partial_\theta \log P_t(\mathbf{x})$ is a function on the same space as the *diffusion* score $\nabla_{\mathbf{x}} \log P_t(\mathbf{x})$. A third, harmless usage, "an arm is scored on the test bundle", means evaluated, and is avoided here.

1.2 The score function and Fisher information

Fix a family p_θ, $\theta \in \Theta$ open in $\mathbb{R}^d$, and write $\mathbb{E}_\theta$ for the expectation under p_θ.

Assumption 1.3 (Regularity)**.** (R1) The support $\{\mathbf{x} : p_\theta(\mathbf{x}) > 0\}$ does not depend on θ. (R2) $\theta \mapsto p_\theta(\mathbf{x})$ is continuously differentiable for every $\mathbf{x}$. (R3) Differentiation with respect to θ may be taken under the integral sign in $\int p_\theta \,\mathrm{d}\mathbf{x}$ and in $\int T(\mathbf{x}) p_\theta(\mathbf{x}) \,\mathrm{d}\mathbf{x}$ for the statistics T considered (a dominating integrable bound on $\|\nabla_\theta p_\theta\|$ locally in θ suffices).

Proposition 1.4 (The score has mean zero)**.** *Under Assumption 1.3,* $\mathbb{E}_\theta[\mathbf{s}_\theta(\mathbf{x})] = \mathbf{0}$.

Proof. $\mathbb{E}_\theta[\mathbf{s}_\theta] = \int \frac{\nabla_\theta p_\theta}{p_\theta} p_\theta \,\mathrm{d}\mathbf{x} = \int \nabla_\theta p_\theta \,\mathrm{d}\mathbf{x} = \nabla_\theta \int p_\theta \,\mathrm{d}\mathbf{x} = \nabla_\theta 1 = \mathbf{0}$, using (R1) so that the integrand is defined on the support and (R3) to exchange ∇_θ and $\int$. □

Definition 1.5 (Fisher information)**.** The **Fisher information matrix** of the family at θ is the covariance of the score,

$$\mathbf{I}(\theta) = \mathbb{E}_\theta\big[\mathbf{s}_\theta(\mathbf{x})\mathbf{s}_\theta(\mathbf{x})^\top\big], \qquad I_{jk}(\theta) = \mathbb{E}_\theta\big[\partial_j \log p_\theta \, \partial_k \log p_\theta\big]. \tag{1.3}$$

For a scalar parameter it is the number $I(\theta) = \mathbb{E}_\theta[(\partial_\theta \log p_\theta)^2]$.

The paper's $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ is this definition for the family $\{P_t^\theta\}$ of noised laws at fixed t, one sequence being one observation.

Proposition 1.6 (Four properties)**.** *Under Assumption 1.3, with second derivatives where stated:*

(i) (second-derivative form) if $\theta \mapsto p_\theta$ is twice differentiable and (R3) holds for the second derivatives, $I_{jk}(\theta) = -\mathbb{E}_\theta[\partial_j \partial_k \log p_\theta]$;

(ii) (additivity) for M independent observations with a common θ the information is $M\,\mathbf{I}(\theta)$; for independent observations from different families it is the sum of their informations;

(iii) (reparametrisation) if $\theta = g(\psi)$ with g differentiable and Jacobian $\mathbf{J} = \partial\theta/\partial\psi$, then $\mathbf{I}_\psi(\psi) = \mathbf{J}^\top \mathbf{I}_\theta(g(\psi))\mathbf{J}$; for scalars, $I_\psi = (g'(\psi))^2 I_\theta$;

(iv) (data processing) if $\mathbf{y} = T(\mathbf{x})$ is a statistic, the information carried by $\mathbf{y}$ about θ is at most that carried by $\mathbf{x}$, in the Loewner order.

Proof. (i) Differentiate $\partial_j \log p = \partial_j p / p$ once more: $\partial_j \partial_k \log p = \partial_j \partial_k p / p - (\partial_j p)(\partial_k p)/p^2$, and $\mathbb{E}_\theta[\partial_j \partial_k p / p] = \int \partial_j \partial_k p \,\mathrm{d}\mathbf{x} = \partial_j \partial_k 1 = 0$ by (R3); the second term's expectation is $-I_{jk}$. (ii) The log-likelihood of independent observations is a sum, so the score is a sum of independent mean-zero vectors and its covariance is the sum of their covariances. (iii) By the chain rule $\nabla_\psi \log p = \mathbf{J}^\top \nabla_\theta \log p$, and covariances transform as $\mathbf{J}^\top \operatorname{Cov} \mathbf{J}$. (iv) Write $p_\theta(\mathbf{x}) = p_\theta(\mathbf{y})\, p_\theta(\mathbf{x} \mid \mathbf{y})$; the score of $\mathbf{x}$ is the score of $\mathbf{y}$ plus the score of the conditional law, and the two are uncorrelated because the conditional score has conditional mean zero given $\mathbf{y}$ (Proposition 1.4 applied to the conditional family); so $\mathbf{I}_\mathbf{x} = \mathbf{I}_\mathbf{y} + \mathbb{E}[\mathbf{I}_{\mathbf{x}|\mathbf{y}}] \succeq \mathbf{I}_\mathbf{y}$. □

Why it matters here. Three of these are used by the paper without comment. Additivity is why the bound for M sequences is $1/(MI_t)$. Reparametrisation is how the paper's Cramér–Rao reference for the excess kurtosis is obtained "through the chain rule at the Laplace member of the one-parameter shape family": the information about β is computed, and the information about $\gamma_2 = \kappa_4/q^2$ as a function of β is $I_\beta / (\,\mathrm{d}\gamma_2/\,\mathrm{d}\beta)^2$, so the reference standard deviation of an unbiased estimator of γ_2 is $|\,\mathrm{d}\gamma_2/\,\mathrm{d}\beta| / \sqrt{M I_\beta}$. Data processing is why the information of the noised sequence is at most that of the clean one, and why the "clean-data floor" of the paper's Figure 3 is a floor.

1.3 The Cramér–Rao bound

Theorem 1.7 (Cramér–Rao, scalar)**.** *Let θ be scalar, let Assumption 1.3 hold, let $0 < I(\theta) < \infty$, and let $T(\mathbf{x})$ be an estimator with $\mathbb{E}_\theta[T^2] < \infty$ and bias $b(\theta) = \mathbb{E}_\theta[T] - \theta$ differentiable with (R3). Then*

$$\operatorname{Var}_\theta(T) \;\geq\; \frac{\big(1 + b'(\theta)\big)^2}{I(\theta)}, \qquad \text{and in particular} \qquad \operatorname{Var}_\theta(T) \geq \frac{1}{I(\theta)} \text{ if } T \text{ is unbiased.} \tag{1.4}$$

Proof. Compute the covariance of T with the score. Since the score has mean zero,

$$\operatorname{Cov}_\theta(T, s_\theta) = \mathbb{E}_\theta[T\, s_\theta] = \int T(\mathbf{x})\, \frac{\partial_\theta p_\theta(\mathbf{x})}{p_\theta(\mathbf{x})}\, p_\theta(\mathbf{x})\, \mathrm{d}\mathbf{x} = \partial_\theta \int T(\mathbf{x}) p_\theta(\mathbf{x})\, \mathrm{d}\mathbf{x} = \partial_\theta \mathbb{E}_\theta[T] = 1 + b'(\theta),$$

using (R3) for the exchange. The Cauchy–Schwarz inequality for covariances, $\operatorname{Cov}(T, s)^2 \leq \operatorname{Var}(T) \operatorname{Var}(s)$, with $\operatorname{Var}_\theta(s_\theta) = I(\theta)$, gives $(1 + b'(\theta))^2 \leq \operatorname{Var}_\theta(T)\, I(\theta)$, which is Equation (1.4). □

Equality in Cauchy–Schwarz holds if and only if $T - \mathbb{E}T$ is a multiple of the score, $T(\mathbf{x}) - \theta = c(\theta)\, s_\theta(\mathbf{x})$; an estimator that achieves the bound at every θ is called **efficient**, and exists only for exponential families in their mean parameterisation. The bound is a property of the family, not of any estimator; that is why the paper calls it a *reference* and not a target.

Theorem 1.8 (Cramér–Rao, vector)**.** *Let $\theta \in \mathbb{R}^d$, let Assumption 1.3 hold with $\mathbf{I}(\theta)$ finite and non-singular, and let $\mathbf{T}(\mathbf{x}) \in \mathbb{R}^d$ be unbiased for θ with finite second moments. Then $\operatorname{Cov}_\theta(\mathbf{T}) \succeq \mathbf{I}(\theta)^{-1}$ in the Loewner order, i.e. $\mathbf{v}^\top \operatorname{Cov}_\theta(\mathbf{T}) \mathbf{v} \geq \mathbf{v}^\top \mathbf{I}(\theta)^{-1} \mathbf{v}$ for every $\mathbf{v}$.*

Proof. As in the scalar case, $\operatorname{Cov}_\theta(\mathbf{T}, \mathbf{s}_\theta) = \mathbb{E}_\theta[\mathbf{T}\mathbf{s}_\theta^\top] = \nabla_\theta \mathbb{E}_\theta[\mathbf{T}]^\top = \mathbf{I}$. The joint covariance of the stacked vector $(\mathbf{T}, \mathbf{s}_\theta)$ is $\left(\begin{smallmatrix} \operatorname{Cov}(\mathbf{T}) & \mathbf{I} \\ \mathbf{I} & \mathbf{I} \end{smallmatrix}\right) \succeq 0$, and a block matrix $\left(\begin{smallmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{C} \end{smallmatrix}\right)$ with $\mathbf{C} \succ 0$ is positive semidefinite if and only if its Schur complement $\mathbf{A} - \mathbf{B}\mathbf{C}^{-1}\mathbf{B}^\top \succeq 0$; here that is $\operatorname{Cov}(\mathbf{T}) - \mathbf{I}^{-1} \succeq 0$. □

1.4 Nuisance parameters and the efficient information

Partition $\theta = (\psi, \eta)$ into a coordinate of interest ψ (scalar, say) and nuisance parameters η, and partition the information accordingly, $\mathbf{I} = \left(\begin{smallmatrix} I_{\psi\psi} & \mathbf{I}_{\psi\eta} \\ \mathbf{I}_{\eta\psi} & \mathbf{I}_{\eta\eta} \end{smallmatrix}\right)$. By Theorem 1.8 the variance of an unbiased estimator of ψ that must also estimate η is bounded below by $(\mathbf{I}^{-1})_{\psi\psi}$, and by the block inverse formula

$$(\mathbf{I}^{-1})_{\psi\psi} = \frac{1}{I_\psi^{\mathrm{eff}}}, \qquad I_\psi^{\mathrm{eff}} = I_{\psi\psi} - \mathbf{I}_{\psi\eta} \mathbf{I}_{\eta\eta}^{-1} \mathbf{I}_{\eta\psi}, \tag{1.5}$$

the Schur complement. I_ψ^{eff} is the **efficient information** about ψ: the information that survives after the nuisance has been estimated. It is at most $I_{\psi\psi}$, with equality if and only if the scores of ψ and of η are uncorrelated. The ratio $I_{\psi\psi} / I_\psi^{\mathrm{eff}} \geq 1$ is the **nuisance cost**, the quantity `summarise` in `src/markov_score/information.py` reports. Equivalently, I_ψ^{eff} is the variance of the **efficient score**, the residual of the ψ-score after its linear regression on the η-score.

Proposition 1.9 (Nuisance at another Hermite degree)**.** *Suppose that, as $t \to \infty$ and with $\alpha = e^{-t}$, the scores of two coordinates of the noised family have expansions $s_\psi = \alpha^m A(\mathbf{x}) + \alpha^{m+2} A_2(\mathbf{x}) + \dots$ and $s_\eta = \alpha^{m'} B(\mathbf{x}) + \alpha^{m'+2} B_2(\mathbf{x}) + \dots$ with A, B polynomials that are γ-orthogonal, $\mathbb{E}_\gamma[AB] = 0$, and with $I_{\psi\psi} = \alpha^{2m} \mathbb{E}_\gamma[A^2](1 + O(\alpha^2))$, $I_{\eta\eta} = \alpha^{2m'} \mathbb{E}_\gamma[B^2](1 + O(\alpha^2))$, $\mathbb{E}_\gamma[B^2] > 0$, and $P_t = \gamma(1 + O(\alpha^2))$ on the relevant moments. Then $I_{\psi\eta} = O(\alpha^{m+m'+2})$ and*

$$I_\psi^{\mathrm{eff}} = I_{\psi\psi}(1 + O(\alpha^4)) = \alpha^{2m} \mathbb{E}_\gamma[A^2](1 + O(\alpha^2)) : \tag{1.6}$$

the efficient information has the same rate and the same leading constant as the raw information. If instead $m = m'$ and $\mathbb{E}_\gamma[AB] \neq 0$, the correction is of the same order as $I_{\psi\psi}$ and the efficient information is smaller by an $O(1)$ factor.

Proof. $I_{\psi\eta} = \mathbb{E}_{P_t}[s_\psi s_\eta] = \alpha^{m+m'}\mathbb{E}_{P_t}[AB] + O(\alpha^{m+m'+2})$, and $\mathbb{E}_{P_t}[AB] = \mathbb{E}_\gamma[AB] + O(\alpha^2) = O(\alpha^2)$. Hence $I_{\psi\eta}^2/I_{\eta\eta} = O(\alpha^{2m+2m'+4})/\Theta(\alpha^{2m'}) = O(\alpha^{2m+4})$, which is $O(\alpha^4)$ relative to $I_{\psi\psi} = \Theta(\alpha^{2m})$. In the same-degree case the cross term is $\alpha^{2m}\mathbb{E}_\gamma[AB] \neq 0$ at leading order and the Schur complement subtracts a fixed fraction. □

Why it matters here. This is the paper's sentence "the leading orders of a degree-2 and a degree-4 coordinate do not couple, because He_2 and He_4 are γ-orthogonal, so the efficient information of a shape coordinate after lower-degree nuisance coordinates are estimated has the same rate and constant as its raw information; a nuisance of the same degree costs an $O(1)$ factor". For the generalised-Gaussian chain, $\psi = \beta$ has $m = 4$ with A a degree-four Hermite polynomial, and $\eta = (\rho, q)$ have $m' = 2$ with B a degree-two one. The proposition says the e^{-8t} law and the constant $\frac{1}{24}\|\partial_\beta \boldsymbol{\kappa}_4\|_F^2$ of the paper's Theorem 2 hold for the efficient information that the paper's Figure 2b actually plots, not only for the diagonal entry. It also says why the eight-component mixture's shape coordinates, which are all of degree four, cost each other an $O(1)$ factor, one reason the paper calls its one-parameter reference a yardstick.

The paper measures the information matrix of (ρ, q, β) by the **outer-product estimator**: with per-sequence scores $\mathbf{g}_\mu = \nabla_\theta \log P_t(\mathbf{x}^\mu)$ at the true θ, $\hat{\mathbf{I}} = \frac{1}{M}\sum_\mu \mathbf{g}_\mu \mathbf{g}_\mu^\top$, which is unbiased for $\mathbf{I}$ by Equation (1.3) (it is not the observed information $-\nabla^2 \log p$, which would need second derivatives). The function `outer_product_information` of `src/markov_score/information.py` forms it, and `efficient_information` inverts it coordinate by coordinate as in Equation (1.5). Its own sampling error is what limits the paper's chain verification of the shape constant; the scalar constant is verified by quadrature instead (E21).

1.5 The Gaussian family in closed form

Proposition 1.10 (Fisher information of a zero-mean Gaussian family)**.** *Let $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}(\theta))$ on $\mathbb{R}^L$ with $\boldsymbol{\Sigma}(\theta)$ positive definite and differentiable in $\theta \in \mathbb{R}^d$. Then*

$$I_{jk}(\theta) = \tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma}\,\boldsymbol{\Sigma}^{-1}\partial_k\boldsymbol{\Sigma}). \tag{1.7}$$

Proof. The log-density is $\log p_\theta(\mathbf{x}) = -\frac{L}{2}\log 2\pi - \frac{1}{2}\log\det\boldsymbol{\Sigma} - \frac{1}{2}\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}\mathbf{x}$. Using $\partial_j \log\det\boldsymbol{\Sigma} = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma})$ and $\partial_j\boldsymbol{\Sigma}^{-1} = -\boldsymbol{\Sigma}^{-1}(\partial_j\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}$,

$$\partial_j \log p_\theta(\mathbf{x}) = -\tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma}) + \tfrac{1}{2}\mathbf{x}^\top\boldsymbol{\Sigma}^{-1}(\partial_j\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}\mathbf{x} = \tfrac{1}{2}\big(\mathbf{x}^\top\mathbf{B}_j\mathbf{x} - \mathbb{E}[\mathbf{x}^\top\mathbf{B}_j\mathbf{x}]\big),$$

with $\mathbf{B}_j = \boldsymbol{\Sigma}^{-1}(\partial_j\boldsymbol{\Sigma})\boldsymbol{\Sigma}^{-1}$ symmetric, since $\mathbb{E}[\mathbf{x}^\top\mathbf{B}\mathbf{x}] = \operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma}) = \operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma})$. The information is the covariance of these centred quadratic forms. For $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ and symmetric $\mathbf{B}$, $\mathbf{C}$, Isserlis' theorem (chapter on Gaussian integrals and Hermite polynomials) gives $\mathbb{E}[\mathbf{x}^\top\mathbf{B}\mathbf{x}\,\mathbf{x}^\top\mathbf{C}\mathbf{x}] = \operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma})\operatorname{tr}(\mathbf{C}\boldsymbol{\Sigma}) + 2\operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma}\mathbf{C}\boldsymbol{\Sigma})$, so $\operatorname{Cov}(\mathbf{x}^\top\mathbf{B}\mathbf{x}, \mathbf{x}^\top\mathbf{C}\mathbf{x}) = 2\operatorname{tr}(\mathbf{B}\boldsymbol{\Sigma}\mathbf{C}\boldsymbol{\Sigma})$. With the factors $\frac{1}{2}\cdot\frac{1}{2}$, $I_{jk} = \frac{1}{4}\cdot 2\operatorname{tr}(\mathbf{B}_j\boldsymbol{\Sigma}\mathbf{B}_k\boldsymbol{\Sigma}) = \frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}^{-1}\partial_j\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1}\partial_k\boldsymbol{\Sigma})$. □

The noised Gaussian chain. For the Gaussian chain of the chapter on Markov chains, $\mathbf{x}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_t(\theta))$ exactly, with $\boldsymbol{\Sigma}_t = e^{-2t}\boldsymbol{\Sigma}_0(\theta) + \Delta_t\mathbf{I}$ and $\theta = (\rho, q)$, so $\partial_\theta\boldsymbol{\Sigma}_t = e^{-2t}\partial_\theta\boldsymbol{\Sigma}_0$ and Equation (1.7) gives the paper's closed form

$$I_t(\theta) = \tfrac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t\,\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_t) = \tfrac{1}{2}e^{-4t}\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_0\,\boldsymbol{\Sigma}_t^{-1}\partial_\theta\boldsymbol{\Sigma}_0), \tag{1.8}$$

exact at every t and every ρ; this is the curve of the paper's Figure 2a and the function `gaussian_chain_fisher` of `src/markov_score/theory.py`, which evaluates $\partial_\theta\boldsymbol{\Sigma}_0$ by central differences. Its large-t limit is immediate: $\boldsymbol{\Sigma}_t \to \mathbf{I}$, so

$$e^{4t}I_t(\theta) \;\longrightarrow\; \tfrac{1}{2}\operatorname{tr}(\partial_\theta\boldsymbol{\Sigma}_0\,\partial_\theta\boldsymbol{\Sigma}_0) = \tfrac{1}{2}\|\partial_\theta\boldsymbol{\Sigma}_0\|_F^2, \tag{1.9}$$

the constant of the paper's equation for a covariance coordinate, checked in Lean as target 3 (`lean/README.md`). The approach to the limit is governed by the spectrum of $\mathbf{\Sigma}_0$: writing $\mathbf{\Sigma}_t = \mathbf{I} + e^{-2t}(\mathbf{\Sigma}_0 - \mathbf{I})$, the Neumann series $\mathbf{\Sigma}_t^{-1} = \mathbf{I} - e^{-2t}(\mathbf{\Sigma}_0 - \mathbf{I}) + e^{-4t}(\mathbf{\Sigma}_0 - \mathbf{I})^2 - \dots$ converges in operator norm exactly when $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0 - \mathbf{I}) < 1$, i.e. $e^{2t} > \lambda_{\max}(\mathbf{\Sigma}_0) - 1$, and its first correction is of relative size $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0)$: this is the origin of the paper's $\epsilon_t = O(e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0))$ and of the crossover $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ below which the measured slope is shallower than -4. The same formula gives the off-diagonal entry $I_{\rho q}$, hence the correlation between the two scores and the efficient information of each, which the paper's Figure 2a uses; the efficient slope is already close to -4 below t_* where the raw diagonal is not, because the nuisance coupling of ρ with q is strong there.

1.6 Local asymptotic normality and the Hájek–Le Cam theorem

Fisher information bounds the variance of *unbiased* estimators. Nothing in the Cramér–Rao bound stops a biased estimator from having a smaller mean squared error at some θ (the constant estimator has zero variance). The statement that $1/I$ is nevertheless the right yardstick for *every* estimator is asymptotic and local, and it is the theorem the paper's corollary on sample complexity invokes.

In plain words. Take M independent sequences from p_θ and look at parameter values within a $1/\sqrt{M}$ neighbourhood of θ, the scale on which the data can distinguish them. As $M \to \infty$ the whole experiment, restricted to that neighbourhood, looks like observing one Gaussian vector with mean $\mathbf{h}$ (the rescaled parameter offset) and covariance $\mathbf{I}(\theta)^{-1}$. In that Gaussian experiment no estimator of $\mathbf{h}$ has worst-case risk below that of the observation itself, whose covariance is $\mathbf{I}^{-1}$. So no estimator sequence can beat $\mathbf{I}(\theta)^{-1}/M$ uniformly over a shrinking neighbourhood, however biased, and the maximum-likelihood estimator, which becomes that Gaussian observation, attains it.

Definition 1.11 (Local asymptotic normality)**.** The family is **locally asymptotically normal** (LAN) at θ with information $\mathbf{I}(\theta)$ if, for every $\mathbf{h} \in \mathbb{R}^d$, the log-likelihood ratio of M independent observations satisfies

$$\log \prod_{\mu=1}^{M} \frac{p_{\theta+\mathbf{h}/\sqrt{M}}(\mathbf{x}^\mu)}{p_\theta(\mathbf{x}^\mu)} = \mathbf{h}^\top \mathbf{\Delta}_M - \tfrac{1}{2}\mathbf{h}^\top \mathbf{I}(\theta)\mathbf{h} + o_p(1), \qquad \mathbf{\Delta}_M = \frac{1}{\sqrt{M}}\sum_\mu \mathbf{s}_\theta(\mathbf{x}^\mu) \Rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}(\theta))\,, \tag{1.10}$$

under p_θ. A sufficient condition is that the family be **differentiable in quadratic mean** at θ: $\sqrt{p_{\theta+\mathbf{h}}} = \sqrt{p_\theta} + \frac{1}{2}\mathbf{h}^\top \mathbf{s}_\theta \sqrt{p_\theta} + r_\mathbf{h}$ with $\int r_\mathbf{h}^2 = o(|\mathbf{h}|^2)$; a continuously differentiable log-density with continuous finite information suffices for that.

Theorem 1.12 (Hájek–Le Cam local asymptotic minimax)**.** *Let the family be LAN at θ with non-singular $\mathbf{I}(\theta)$, let $\ell : \mathbb{R}^d \to [0, \infty)$ be bowl-shaped (its sublevel sets convex and symmetric), and let $\mathbf{T}_M$ be any sequence of estimators. Then*

$$\lim_{c\to\infty} \liminf_{M\to\infty} \sup_{|\mathbf{h}|\le c} \mathbb{E}_{\theta+\mathbf{h}/\sqrt{M}}\, \ell\big(\sqrt{M}(\mathbf{T}_M - \theta - \mathbf{h}/\sqrt{M})\big) \;\ge\; \mathbb{E}\,\ell(\mathbf{Z}), \qquad \mathbf{Z} \sim \mathcal{N}\Big(\mathbf{0}, \mathbf{I}(\theta)^{-1}\Big)\,. \tag{1.11}$$

With the quadratic loss $\ell(\mathbf{u}) = (v^\top \mathbf{u})^2$ for a coordinate direction, $\mathbb{E}\ell(\mathbf{Z}) = (\mathbf{I}^{-1})_{\psi\psi} = 1/I_\psi^{\mathrm{eff}}$, and Equation (1.11) reads: for any estimator sequence $\hat{\psi}_M$,

$$\lim_{c\to\infty} \liminf_{M\to\infty} \sup_{|\theta'-\theta|\le c/\sqrt{M}} M\,\mathbb{E}_{\theta'}(\hat{\psi}_M - \psi')^2 \;\ge\; \frac{1}{I_\psi^{\mathrm{eff}}(\theta)}. \tag{1.12}$$

This is the statement the paper's corollary uses, with I^{eff} the efficient information of the paper's general-order theorem at fixed t, so that the right-hand side is $\Theta(e^{2mt})$ for a coordinate of lowest

Hermite degree m. Two features of the exact statement deserve attention. The supremum over a shrinking neighbourhood is essential: without it a superefficient estimator can beat $1/I$ at isolated points. And the outer limit $c \to \infty$ is part of the theorem; for a fixed c the local minimax risk of the limiting Gaussian experiment over $|\mathbf{h}| \le c$ is strictly below $\mathbb{E}\ell(\mathbf{Z})$ (a bounded normal-mean problem has a shrinkage estimator with smaller worst-case risk), and the bound is approached as c grows. The paper's corollary writes the bound with a fixed c and without the outer limit; read literally that is slightly stronger than Equation (1.12), and the intended statement is Equation (1.12).

What "maximum likelihood attains it" means. Under the regularity that gives LAN plus conditions for consistency, the maximum-likelihood estimator satisfies $\sqrt{M}(\hat{\theta}_M - \theta) \Rightarrow \mathcal{N}(\mathbf{0}, \mathbf{I}(\theta)^{-1})$ under p_θ, locally uniformly, and is **regular** in the sense of the convolution theorem; so its rescaled risk converges to $\mathbb{E}\ell(\mathbf{Z})$ and equality holds in Equation (1.11). Three things this does *not* say, all named by the paper. (i) It is a statement at fixed t as $M \to \infty$; the LAN approximation needs $M \gg 1/I_t$, i.e. $M \gg e^{2mt}$, so the limits $M \to \infty$ and $t \to \infty$ do not commute and nothing is claimed uniformly in t. (ii) It is a statement about *local* precision, on the scale $1/\sqrt{MI_t}$; once that scale exceeds the range of a bounded coordinate (the excess kurtosis of the mixture family is bounded below by -2 and the paper's shape coordinate has range 3), the neighbourhood $|\theta' - \theta| \le c/\sqrt{M}$ is no longer inside the parameter space and the statement is empty; the paper reads its estimator's collapse at that level as the information running out, not as a failure of the estimator. (iii) It says nothing about an optimiser stopped at a finite budget, nor about an estimator whose iteration count is selected on validation.

1.7 Why "at the bound" is a yardstick and not an efficiency theorem

The paper compares, level by level, the dispersion across sixteen replicates of its fitted correlation and fitted excess kurtosis with the Cramér–Rao reference $1/\sqrt{MI_t^{\mathrm{eff}}}$. Four facts make that comparison a yardstick.

1. The reference is computed in a *one-parameter* shape family (the generalised Gaussian at its Laplace member, with (ρ, q) as lower-degree nuisances that cost nothing to leading order, Proposition 1.9). The estimator fits an *eight-component mixture*, whose shape coordinates are all of degree four and cost each other $O(1)$ factors. The two families share the Laplace law as a point but not their tangent spaces, so the reference is not the bound for the estimator's own family.

2. The Cramér–Rao bound is for *unbiased* estimators, and Theorem 1.7 shows that a bias with $b'(\theta) < 0$ lowers the bound by $(1 + b')^2$. Early stopping selected on validation is a shrinkage device: it pulls the fitted shape toward the Gaussian innovation as the information falls. A dispersion *below* the unbiased reference, which the paper reports at some levels, is therefore not a contradiction; it is the signature of shrinkage, and the paper reports bias and dispersion separately for that reason.

3. $M = 512$ is fixed and moderate, whereas the theorem is asymptotic. At the levels where $1/\sqrt{MI_t}$ is a sizeable fraction of the coordinate's range, the local statement is empty (Section 1.6, point (ii)).

4. The reported dispersion is a sixteen-replicate standard deviation, whose own relative sampling error is about $1/\sqrt{2 \cdot 15} \approx 18\%$ under normality (chapter on the statistics of the experiments); "within the sampling error of a sixteen-replicate standard deviation" is the most that can be claimed.

The defensible statement, and the one the paper makes, is that at the low levels the fits are unimodal and their dispersion is consistent with the one-parameter reference, that collapse to a Gaussian innovation begins at a stated level, and that it is total at the level where the reference crosses the coordinate's range.

1.8 Fisher's identity for latent-variable models

The paper's estimator maximises the marginal likelihood of the noised sequences, $\mathcal{L}(\theta) = \sum_\mu \log P_t[t_\mu, \theta](\mathbf{x}^\mu)$, where each term is an integral over the unobserved clean chain. Its gradient is obtained without differentiating through the integral, by an identity due to Fisher.

Proposition 1.13 (Fisher's identity)**.** *Let $p_\theta(\mathbf{a}, \mathbf{x})$ be a joint density of a latent $\mathbf{a}$ and an observation $\mathbf{x}$, with marginal $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{a}, \mathbf{x})\,\mathrm{d}\mathbf{a}$, differentiable in θ with (R3). Then*

$$\nabla_\theta \log p_\theta(\mathbf{x}) = \mathbb{E}_\theta\big[\nabla_\theta \log p_\theta(\mathbf{a}, \mathbf{x}) \,\big|\, \mathbf{x}\big], \tag{1.13}$$

the posterior expectation of the complete-data score.

Proof.

$$\nabla_\theta \log p_\theta(\mathbf{x}) = \frac{\nabla_\theta \int p_\theta(\mathbf{a}, \mathbf{x})\,\mathrm{d}\mathbf{a}}{p_\theta(\mathbf{x})} = \int \frac{\nabla_\theta p_\theta(\mathbf{a}, \mathbf{x})}{p_\theta(\mathbf{a}, \mathbf{x})} \cdot \frac{p_\theta(\mathbf{a}, \mathbf{x})}{p_\theta(\mathbf{x})}\,\mathrm{d}\mathbf{a} = \int \nabla_\theta \log p_\theta(\mathbf{a}, \mathbf{x})\, p_\theta(\mathbf{a} \mid \mathbf{x})\,\mathrm{d}\mathbf{a},$$

using (R3) to move the gradient inside the integral and $p_\theta(\mathbf{a} \mid \mathbf{x}) = p_\theta(\mathbf{a}, \mathbf{x})/p_\theta(\mathbf{x})$. □

Equivalently, with the EM auxiliary function $\mathcal{Q}(\theta \mid \theta') = \mathbb{E}_{\theta'}[\log p_\theta(\mathbf{a}, \mathbf{x}) \mid \mathbf{x}]$, Equation (1.13) says $\nabla_\theta \mathcal{Q}(\theta \mid \theta')|_{\theta=\theta'} = \nabla_\theta \log p_{\theta'}(\mathbf{x})$: the gradient of the EM surrogate at the current point is the gradient of the marginal log-likelihood, which is why EM's ascent directions are likelihood ascent directions (chapter on maximum likelihood and EM).

For the chain. The complete-data density is **??** of the chapter on Markov chains times the channel, $p_\theta(\mathbf{a}, \mathbf{x}) = \mu(a_1) \prod_{i\geq 2} K_\theta(a_i \mid a_{i-1}) \prod_i \ell_t(x_i \mid a_i)$, and only the transition factors depend on θ. So

$$\begin{aligned} \nabla_\theta \log p_\theta(\mathbf{x}) &= \sum_{i\geq 2} \mathbb{E}_\theta\big[\nabla_\theta \log K_\theta(a_i \mid a_{i-1}) \,\big|\, \mathbf{x}\big] \\ &= \sum_{i\geq 2} \iint \nabla_\theta \log K_\theta(a' \mid a)\, p_\theta(a_i = a', a_{i-1} = a \mid \mathbf{x})\,\mathrm{d}a\,\mathrm{d}a', \end{aligned} \tag{1.14}$$

a sum of integrals against the **pairwise posterior marginals** of consecutive sites, which the forward–backward recursion returns in the same pass that computes the site marginals. On the grid, with $\Xi_{kl} = \sum_\mu \sum_{i\geq 2} p(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu)$ the expected transition mass (child first, parent second), the gradient of the whole marginal log-likelihood is the inner product $\langle \Xi, \nabla_\theta \log K_\theta \rangle$ of the paper's equation (4). The same Ξ is all the M-step of EM needs. The identity is verified in the paper against finite differences of the grid evidence to 10^{-9}, and, independently of the recursion, `tests/test_fisher_bruteforce.py` enumerates every latent path on a five-point grid and checks the evidence, the orientation of Ξ (a transposed Ξ paired with a transposed kernel would pass a finite-difference check), and the gradient against a finite difference of the enumerated evidence. The per-sequence scores $\mathbf{g}_\mu$ that feed the outer-product information estimator of Section 1.4 are computed by the batched version of the same identity in `src/markov_score/information.py` (`per_chain_scores`), one term per edge, without materialising a per-sequence $N_\mathrm{g} \times N_\mathrm{g}$ matrix.

1.9 Worked example: the variance of one noised Gaussian coordinate

Let $x \sim \mathcal{N}(0, \theta)$, one scalar observation, $\theta > 0$. The log-density is $-\frac{1}{2}\log(2\pi\theta) - x^2/(2\theta)$, the score is $-1/(2\theta) + x^2/(2\theta^2)$, and since $\mathbb{E}x^2 = \theta$ and $\mathrm{Var}(x^2) = 2\theta^2$,

$$I(\theta) = \mathrm{Var}\Big(\frac{x^2}{2\theta^2}\Big) = \frac{2\theta^2}{4\theta^4} = \frac{1}{2\theta^2},$$

which agrees with Equation (1.7): $\frac{1}{2}\,\mathrm{tr}(\theta^{-1}\cdot 1\cdot\theta^{-1}\cdot 1) = 1/(2\theta^2)$. The Cramér–Rao bound for M unbiased observations is $2\theta^2/M$, attained by the sample second moment $\frac{1}{M}\sum x_\mu^2$, whose variance is exactly $2\theta^2/M$: an efficient estimator, as the exponential-family remark after Theorem 1.7 predicts.

Now noise the coordinate: $x_t = e^{-t}a + \sqrt{\Delta_t}z$ with $a \sim \mathcal{N}(0,\theta)$, so $x_t \sim \mathcal{N}(0, v_t)$ with $v_t = e^{-2t}\theta + 1 - e^{-2t}$ and $\partial_\theta v_t = e^{-2t}$. By Equation (1.7),

$$I_t(\theta) = \frac{(\partial_\theta v_t)^2}{2v_t^2} = \frac{e^{-4t}}{2\big(1 + e^{-2t}(\theta - 1)\big)^2}.$$

At $\theta = 1$ the denominator is exactly 1 at every t, so $I_t(1) = \frac{1}{2}e^{-4t}$ *exactly*: the constant $\frac{1}{2}\|\partial_\theta \mathbf{\Sigma}_0\|_F^2 = \frac{1}{2}$ of the paper's covariance-coordinate law, with no correction, because a unit-variance coordinate keeps unit variance under the channel. This is the "$I_t e^{4t}$ equal to one half" that the paper's Appendix E checks for the variance coordinate of the scalar law, and that `tests/test_theory_hermite.py` checks by quadrature for the Laplace law as well (the paper's Theorem 2 says the constant is the same for every law with an exponential moment). At $\theta \neq 1$ the factor $(1+e^{-2t}(\theta-1))^{-2}$ is the scalar version of the crossover: for $\theta = 5$ the local log-slope of I_t is $-4+4\cdot 4e^{-2t}/(1+4e^{-2t})$, about -1.6 at $t = 0.5$, -3.3 at $t = 1.5$ and -4.0 at $t = 3$, with the crossover $\frac{1}{2}\log(\theta-1) = \frac{1}{2}\log 4 \approx 0.69$ at which the correction term equals one. With $M = 512$ sequences at $t = 1$ and $\theta = 1$, the reference standard deviation is $\sqrt{2e^4/512} \approx 0.46$, more than the range of a coordinate confined to an interval of length one: at that level the local theory has nothing to say about such a coordinate, which is the content of point (ii) in Section 1.6.

Where it is used. The paper's Section 3 (the definition of $I_t(\theta)$, the closed-form Gaussian information, the efficient-information sentence, Theorem 3 on the general order, Corollary 4 and the three caveats after it), Section 4 (the Cramér–Rao reference, "a yardstick, not an efficiency theorem", Fisher's identity and equation (4)), Appendix A (the proof of the corollary and the nuisance-projection argument), Appendix C (the finite-difference verification) and Appendix E (E21, the scalar constants). Experiments E01, E06, E12b, E19, E21. Code: `src/markov_score/theory.py` (`gaussian_chain_fisher`, `correlation_information_constant`, `shape_information_constant`), `src/markov_score/information.py` (`per_chain_scores`, `outer_product_information`, `efficient_information`, `summarise`), `src/markov_score/shape_kernel.py` (the shape coordinate and its gradient), `tests/test_fisher_bruteforce.py`, `tests/test_theory_hermite.py`, `tests/test_shape_kernel.py`.

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